

Package ‘defactor’

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Title Tools for Managing and Annotating Data Frame Variables with Labels

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Description Provides a set of functions to efficiently manage, annotate, and retrieve variable labels in data frames. This package allows for dynamic setting and extraction of labels based on patterns, column selections, or custom specifications. It is particularly useful for data cleaning, analysis, and preparation tasks where labeled data is critical for interpretation and reporting.

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Depends R (>= 3.5.0)

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Encoding UTF-8

Roxygen list(markdown = TRUE)

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colInfo

*Get Details for All Columns***Description**

Retrieves labels and details for all columns in the data frame.

Usage

```
colInfo(df, limit = 0, rep = FALSE)
```

Arguments

df	A data frame.
limit	The maximum number of labels per column (default: 0, which means all labels).
rep	A logical value indicating whether to allow repetitions (default: FALSE).

Value

A data frame with variable names, labels, category values, and category labels.

Examples

```
library(defactor)
library(data.table)
# Create the data table
df <- data.table(
  C_COW_ALPHA = c("BOS", "BOS", "BOS", "BOS", "BOS"),
  COW = c(346, 346, 346, 346, 346),
  B_COUNTRY_ALPHA = c("BIH", "BIH", "BIH", "BIH", "BIH"),
  V2 = c(70, 70, 70, 70, 70),
  V2A = c(70, 70, 70, 70, 70),
  V3 = c(1, 2, 3, 4, 5),
  V4 = c(1, 1, 1, 1, 1),
  V5 = c(1, 1, 1, 1, 1),
  V6 = c(1, 2, 2, 1, 2)
)

# Add column labels as attributes
attr(df$COW, "label") <- "column1"
attr(df$V2, "label") <- "column2"
attr(df$V2A, "label") <- "column3"
attr(df$V4, "label") <- "column4"
attr(df$V5, "label") <- "column5"
attr(df$V6, "label") <- "column6"

# Add value labels as attributes
attr(df$COW, "labels") <- c("Bosnia and Herzegovina" = 346)
attr(df$V2, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V2A, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V4, "labels") <- c("Very important" = 1)
attr(df$V5, "labels") <- c("Very important" = 1)
attr(df$V6, "labels") <- c("Very important" = 1, "Rather important" = 2)
colInfo(df, limit = 1)
```

getLbl	<i>Get Label for a Column</i>
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Description

Retrieves the labels of a specified column in a data frame.

Usage

```
getLbl(df, col, suffix = "_labels", limit = 0)
```

Arguments

df	A data frame containing the column.
col	The column name for which labels are to be retrieved.
suffix	A string to append to the label column name (default: "_labels").
limit	The maximum number of labels to return (default: 0, which means all labels).

Details

The function retrieves the labels attribute of a factor variable and uses it to create a new column with descriptive labels. The new column name is either specified by the bycol parameter or dynamically generated using the suffix parameter.

Value

A data frame with the column values and their corresponding labels.

Examples

```
# Create a sample data frame with a factor column
# Load the package
library(defactor)

#Load the required package
library(data.table)
# Create the data table
df <- data.table(
  C_COW_ALPHA = c("BOS", "BOS", "BOS", "BOS", "BOS"),
  COW = c(346, 346, 346, 346, 346),
  B_COUNTRY_ALPHA = c("BIH", "BIH", "BIH", "BIH", "BIH"),
  V2 = c(70, 70, 70, 70, 70),
  V2A = c(70, 70, 70, 70, 70),
  V3 = c(1, 2, 3, 4, 5),
  V4 = c(1, 1, 1, 1, 1),
  V5 = c(1, 1, 1, 1, 1),
  V6 = c(1, 2, 2, 1, 2)
)

# Add column labels as attributes
attr(df$COW, "label") <- "column1"
attr(df$V2, "label") <- "column2"
```

```

attr(df$V2A, "label") <- "column3"
attr(df$V4, "label") <- "column4"
attr(df$V5, "label") <- "column5"
attr(df$V6, "label") <- "column6"

# Add value labels as attributes
attr(df$COW, "labels") <- c("Bosnia and Herzegovina" = 346)
attr(df$V2, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V2A, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V4, "labels") <- c("Very important" = 1)
attr(df$V5, "labels") <- c("Very important" = 1)
attr(df$V6, "labels") <- c("Very important" = 1, "Rather important" = 2)
attr(df$V6, "labels") <- c("Very important" = 1, "Rather important" = 2)

```

listVars

*Get Variable Labels***Description**

Retrieves all variable names and their labels from the data frame.

Usage

```
listVars(df)
```

Arguments

df A data frame.

Value

A data frame with variable names and their corresponding labels.

Examples

```

library(defactor)
library(data.table)
# Create the data table
df <- data.table(
  C_COW_ALPHA = c("BOS", "BOS", "BOS", "BOS", "BOS"),
  COW = c(346, 346, 346, 346, 346),
  B_COUNTRY_ALPHA = c("BIH", "BIH", "BIH", "BIH", "BIH"),
  V2 = c(70, 70, 70, 70, 70),
  V2A = c(70, 70, 70, 70, 70),
  V3 = c(1, 2, 3, 4, 5),
  V4 = c(1, 1, 1, 1, 1),
  V5 = c(1, 1, 1, 1, 1),
  V6 = c(1, 2, 2, 1, 2)
)

# Add column labels as attributes
attr(df$COW, "label") <- "column1"
attr(df$V2, "label") <- "column2"
attr(df$V2A, "label") <- "column3"

```

```

attr(df$V4, "label") <- "column4"
attr(df$V5, "label") <- "column5"
attr(df$V6, "label") <- "column6"

# Add value labels as attributes
attr(df$COW, "labels") <- c("Bosnia and Herzegovina" = 346)
attr(df$V2, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V2A, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V4, "labels") <- c("Very important" = 1)
attr(df$V5, "labels") <- c("Very important" = 1)
attr(df$V6, "labels") <- c("Very important" = 1, "Rather important" = 2)
listVars(df)

```

setLbl

*Set Label for a Column***Description**

Merges labels into the original data frame for a specified column.

Usage

```
setLbl(df, col, suffix = "_label")
```

Arguments

df	A data frame.
col	The column name for which labels are to be set.
suffix	A string to append to the label column name (default: "_label").

Value

A data frame with the labels merged into the original data frame.

Examples

```

library(defactor)
library(data.table)
# Create the data table
df <- data.table(
  C_COW_ALPHA = c("BOS", "BOS", "BOS", "BOS", "BOS"),
  COW = c(346, 346, 346, 346, 346),
  B_COUNTRY_ALPHA = c("BIH", "BIH", "BIH", "BIH", "BIH"),
  V2 = c(70, 70, 70, 70, 70),
  V2A = c(70, 70, 70, 70, 70),
  V3 = c(1, 2, 3, 4, 5),
  V4 = c(1, 1, 1, 1, 1),
  V5 = c(1, 1, 1, 1, 1),
  V6 = c(1, 2, 2, 1, 2)
)

# Add column labels as attributes
attr(df$COW, "label") <- "column1"

```

```

attr(df$V2, "label") <- "column2"
attr(df$V2A, "label") <- "column3"
attr(df$V4, "label") <- "column4"
attr(df$V5, "label") <- "column5"
attr(df$V6, "label") <- "column6"

# Add value labels as attributes
attr(df$COW, "labels") <- c("Bosnia and Herzegovina" = 346)
attr(df$V2, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V2A, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V4, "labels") <- c("Very important" = 1)
attr(df$V5, "labels") <- c("Very important" = 1)
attr(df$V6, "labels") <- c("Very important" = 1, "Rather important" = 2)
setLbl(df, "V5", suffix = "_label")

```

setLblPat

Set Labels for All Columns by Pattern

Description

Sets labels for all columns in a data frame that match a specified regex pattern.

Usage

```
setLblPat(df, pattern = ".*")
```

Arguments

df	A data frame.
pattern	A regex pattern to match column names (default: ".*" for all columns).

Value

A data frame with labels set for the matching columns.

Examples

```

library(defactor)
library(data.table)
# Create the data table
df <- data.table(
  C_COW_ALPHA = c("BOS", "BOS", "BOS", "BOS", "BOS"),
  COW = c(346, 346, 346, 346, 346),
  B_COUNTRY_ALPHA = c("BIH", "BIH", "BIH", "BIH", "BIH"),
  V2 = c(70, 70, 70, 70, 70),
  V2A = c(70, 70, 70, 70, 70),
  V3 = c(1, 2, 3, 4, 5),
  V4 = c(1, 1, 1, 1, 1),
  V5 = c(1, 1, 1, 1, 1),
  V6 = c(1, 2, 2, 1, 2)
)

# Add column labels as attributes
attr(df$COW, "label") <- "column1"

```

```

attr(df$V2, "label") <- "column2"
attr(df$V2A, "label") <- "column3"
attr(df$V4, "label") <- "column4"
attr(df$V5, "label") <- "column5"
attr(df$V6, "label") <- "column6"

# Add value labels as attributes
attr(df$COW, "labels") <- c("Bosnia and Herzegovina" = 346)
attr(df$V2, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V2A, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V4, "labels") <- c("Very important" = 1)
attr(df$V5, "labels") <- c("Very important" = 1)
attr(df$V6, "labels") <- c("Very important" = 1, "Rather important" = 2)
setLbIPat(df, pattern = "^V5")

```

setlstLbl

Set Labels for All Columns by Selected Names

Description

Sets labels for all columns in a data frame that match a specified list of column names.

Usage

```
setlstLbl(df, columns = c())
```

Arguments

df	A data frame.
columns	A character vector of column names for which labels should be set.

Value

A data frame with labels set for the specified columns.

Examples

```

library(defactor)
library(data.table)
# Create the data table
df <- data.table(
  C_COW_ALPHA = c("BOS", "BOS", "BOS", "BOS", "BOS"),
  COW = c(346, 346, 346, 346, 346),
  B_COUNTRY_ALPHA = c("BIH", "BIH", "BIH", "BIH", "BIH"),
  V2 = c(70, 70, 70, 70, 70),
  V2A = c(70, 70, 70, 70, 70),
  V3 = c(1, 2, 3, 4, 5),
  V4 = c(1, 1, 1, 1, 1),
  V5 = c(1, 1, 1, 1, 1),
  V6 = c(1, 2, 2, 1, 2)
)

# Add column labels as attributes
attr(df$COW, "label") <- "column1"

```

```
attr(df$V2, "label") <- "column2"
attr(df$V2A, "label") <- "column3"
attr(df$V4, "label") <- "column4"
attr(df$V5, "label") <- "column5"
attr(df$V6, "label") <- "column6"

# Add value labels as attributes
attr(df$COW, "labels") <- c("Bosnia and Herzegovina" = 346)
attr(df$V2, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V2A, "labels") <- c("Bosnia Herzegovina" = 70)
attr(df$V4, "labels") <- c("Very important" = 1)
attr(df$V5, "labels") <- c("Very important" = 1)
attr(df$V6, "labels") <- c("Very important" = 1, "Rather important" = 2)
setlstLbl(df, columns = c("V2", "V4"))
```

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